

Kyungtaek Lee

VEHICLE MOTION PLANNING & CONTROL ENGINEER

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PROFESSIONAL SUMMARY

Vehicle motion control and autonomous driving engineer with more than seven years of experience in production automotive engineering. Technical lead and project owner for safety-critical planning and control functions, with end-to-end responsibility spanning function architecture, algorithm development, modeling and simulation, real-time implementation, vehicle integration, calibration, and validation. Expertise includes motion planning, trajectory generation and tracking, vehicle dynamics, integrated chassis control, state estimation, and optimization-based control.

CORE COMPETENCIES

- **Planning & Control:** Motion planning, trajectory generation, path tracking, integrated chassis control, optimization-based control
- **Vehicle Engineering:** Vehicle dynamics, state estimation, actuator coordination, calibration, in-vehicle testing, recorded-data analysis
- **Programming:** C for production software analysis and development; MATLAB; Python
- **Real-Time Development:** MATLAB/Simulink, dSPACE MicroAutoBox II, real-time model build and deployment, rapid control prototyping
- **NVIDIA Platforms:** NVIDIA Jetson AGX; Isaac Sim
- **Simulation & Process:** CarSim, CarMaker, AUTOSAR-aligned development, unit/integration testing; familiarity with ISO 26262, ISO 21448 SOTIF, and Automotive SPICE

PROFESSIONAL EXPERIENCE

Senior Research Engineer | HL Mando

2015.08 - Present

- **Mobility Motion Control (2020 - Present):** Technical lead and project owner for planning and control functions including Integrated Chassis Control, Vehicle Stability Control Assist, Evasive Collision Avoidance, Smart Hitching Assist, Trailer Parking Assist, and Minimum Risk Maneuver.
- Lead function definition and algorithm development through simulation, real-time deployment, system integration, vehicle tuning, scenario-based testing, and measurable vehicle-level evaluation.
- Develop and analyze production software in C; build and deploy MATLAB/Simulink control models to dSPACE MicroAutoBox II; use NVIDIA Jetson AGX for vehicle evaluation.
- **System Design (2019 - 2020):** E-Corner Module system design and analysis.
- **Gear Design (2015 - 2019):** Electric Power Steering, Steer-by-Wire, Shift-by-Wire, and e-Drive systems; durability analysis, optimization, and in-house engineering tool development.

Research Engineer | Samsung Techwin (now Hanwha Aerospace)

2012.08 - 2015.07

- Designed integral helical gears for turbo compressors and performed core-system design and dynamic-system analysis.

Engineering Intern | General Motors Korea

2011.07 - 2011.08

- Supported engine-map tuning and validation.

SELECTED PROJECTS

Trailer Parking Assist & Minimum Risk Maneuver HL Mando	2026 - Present
- Technical lead and project owner for planning and control architecture, simulation, real-time implementation, system integration, and vehicle-level evaluation. - Developing robust maneuver generation and safe fallback behavior for production-oriented automated-driving functions.	
Smart Hitching Assist HL Mando	2025
- Led end-to-end function architecture, planning and control development, real-time deployment, vehicle integration, calibration, and validation. - Delivered a successful customer demonstration; received a Company Special Recognition Award in December 2025.	
Evasive Collision Avoidance HL Mando	2024
- Owned the complete planning and control scope: evasive path generation, path tracking control, and vehicle stability coordination. - Integrated and validated the unified framework through simulation, dSPACE real-time execution, in-vehicle tuning, and safety-critical scenario testing.	
Vehicle Stability Control Assist HL Mando	2024
- Developed a hierarchical integrated chassis controller combining differential braking and semi-active suspension damping. - Published simulation and real-world evaluation results showing approximately 17.4% lower maximum roll angle and 8.7% lower maximum sideslip angle versus respective conventional methods.	

EDUCATION

Ph.D. Candidate, Mechanical Engineering Seoul National University	2023.03 - Present
Interactive and Networked Robotics Laboratory (INRoL), Advisor: Professor Dongjun Lee Research: Supervisory Integrated Chassis Control Using Model Predictive Path Integral Control with Legacy Controllers	
M.S., Mechanical Engineering Seoul National University	2020.09 - 2022.06
Vehicle Dynamics and Control Laboratory (VDCL), Advisor: Professor Kyongsu Yi Thesis: Path Tracking Control of Four-Wheel-Independent-Steering-and-Driving Vehicle Based on Adaptive-Weight Optimal Control	
B.S., Mechanical Engineering Ajou University	2006.03 - 2012.06

SELECTED HONORS

2025.12 Company Special Recognition Award - Smart Hitching Assist development and customer demonstration, HL Mando 2025.11 Outstanding Paper Award (Oral Session), Korean Society of Automotive Engineers 2024.12 Excellence Award - R&D Outstanding Paper, HL Mando 2024.11 Outstanding Paper Award (Poster Session), Korean Society of Automotive Engineers 2024.03 Best Dialogue Award, EVS37 International Electric Vehicle Symposium and Exhibition	
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SELECTED PUBLICATIONS & PATENTS

K. Lee and J. Seol, "Development of Integrated Chassis Control of Semi-Active Suspension with Differential Brake for Vehicle Lateral Stability," World Electric Vehicle Journal, 16(2):91, 2025. K. Lee et al., "Lyapunov-Informed Model Predictive Path Integral Control for Robust Trailer Hitch Assist under Perception Uncertainty," KSAE Annual Conference, 2025. K. Lee et al., "Continuous Curvature Path Planning Based on Bezier Curves for Autonomous Driving in Complex Environments," KSAE Annual Conference, 2024. - Inventor or co-inventor on 19 patent applications spanning automotive chassis, steering, suspension, vehicle-state estimation, trailer assistance, and mechanical systems.	
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